zoo506 past papers solved

final term

1. What damage caused by termites?

SYMPTOMS OF DAMAGE:

They enter through cut ends of setts & feed on the soft tissue. The tunnel is filled with the soil. The termites attack setts, shoots, canes and also stubbles Entire shoot dries up and can be pulled out easily

2. What should be kept in mind while field scouting?

a field scouting must know:

- How to properly identify pests and their damage
- How to get an accurate assessment of pest populations
- A field scouts' objective is to provide an accurate and unbiased objective of pest populations
- What tools and procedures do you use for alfalfa weevil for potato leafhopper, etc.
- Both of these examples are pests on alfalfa; however, the sampling procedure is different.

3. What are cerci?

A pair of antennae like structure (sometimes short sometime long) present at the end of abdomen are known as cerci

4. What is meant by economic threshold?

Economic threshold.

The pest density at which management action should be taken to prevent an increasing pest population from reaching the economic injury.

5. How many types of injurious species are present?

Those insects which attack injurious species probably form the most important group They are divided into two sections:

1-The parasites that live in or on the different stages of other insects, then called hosts

2-The predators that capture and devour other insects

6. Symptoms of red cotton bug?

Dusky cotton bug: Oxycarenus hyalinipennis

Symptom of damage

- Sucks the sap from developing seeds in open bolls and stains the lint black.
- Seeds discolored and shrunken

7. What are attractants?

Attractants Sometimes called attrahents are materials used to lure insects to traps and poisoned baits.

The use of geraniol in Japanese beetle traps, sugar sirups for ant baits, and fermenting sirup solutions for trapping various moths are good examples.

8. What is difference between intelligent cultivation and destruction of vegetable?

1- Intelligent cultivation and use of fertilizers are well as pruning to remove injured or diseased parts will aid in insect control

2- The destruction of vegetable trash and weeds, frequently called clean culture, is also an important factor, especially in the control of hibernating insects

9. What is meant by hopper burn?

A heavy infestation produces symptoms of “hopper burn” i.e., leaves become dry and brown after insect feeding and patches of burned plants are often lodged.

10. Insect borne diseases (5)

Insect Vector born Diseases in Human

- insects that are highly injurious is that of human and animal health and comfort. Some insects, notably mosquitoes and flies, annoy man and other animals by their bites and stings.

- Other insects such as lice have become adapted to more or less continuous life upon animals and cause much discomfort and loss of vitality
- Not only do insects live externally on animals but also internally in the skin tissues, and in a few cases within the flesh and in the alimentary tract
- Besides the mental and bodily loss due to their presence, insects transmit disease organisms from one person or animal to another Just as in the transmission of plant diseases the insect may be an accidental agent or an essential carrier

A few of the more important insect-borne disease in organisms are

- Malaria
- Bubonic plague
- Typhoid
- Typhus
- Yellow fevers.

11. Physical control (5)

The extremes of both temperature and humidity, either high or low, are not only effective barriers in the natural control of insects

But they are also useful in artificial control

Light and electricity are other factors that come under this heading Physical Factor Controlling Insects

1- Heat

2-Cold

3-Moisture

4-Light

5- Electricity

1- Heat

Most insects cease reproduction and soon die at temperatures rom 100 to 1l0°F

2- The temperature of their surroundings should be raised to 120° or higher to kill them with heat artificially within a few hours

3- The length of period for effective treatment is more dependent upon the time to heat the grain.

2- Cold

1. Although low temperatures are not so effective as high ones in killing insects

2. Many insect pests of stored products such as furniture, rugs, clothing and seeds become inactive at 40 to 45°F.

3. Under these conditions infested materials suffer no further damage or increase in infestation

4. And materials free from insects when put into cold storage will remain so.

5. Those insects which hibernate outdoors withstand temperatures far below zero.

3-Moisture

A conspicuous example of the manipulation of moisture to control insects

Is the drying of grain to reduce the likelihood of weevil injury

4- Light

It is well known that many insects are attracted to bright lights

This fact has been used extensively to trap insects, sometimes for the purpose of studying their distribution, sometimes as a control measure

5- Electricity

1. Devices are available for electrocuting insects

   I-Some in the form of electrical window screens for flies

   II-Others as electric light traps for moths and beetles

   2. These are so arranged that an insect striking parallel wires of the device completes an electrical circuit and is killed by the shock.

12. Symptoms of red cotton bug (5)

Red cotton bug:

Symptom of damage

Red stained lint and rotting bolls.

Inner boll wall with warty growth or water-soaked spots.

Young bolls abort and turn dark brown

The bacterium Nematospora gossypii enters the site of injury and stains the fiber.

13. Chemical control (2)

The chemical control of insects includes control not only with insecticides in the true sense but also with various materials used to attract and repel insects without actually killing them. Example are

Attractants

Repellents

14. Scientific name of early shoot borer (2)

Scientific name of early shoot borer is **Trichogramma chilonis**

15. Pest management (2)

The stud of principles and experimental approaches to the science that underpins the development of working integrated pest management systems

Types of insect control

A-NATURAL CONTROL

B-ARTIFICIAL CONTROL

Measures for the artificial control of insects may be divided into six groups:

(1) mechanical control

(2) cultural control

(3) biological control

(4) legislative control

(5) physical control

(6) chemical control.

16. Strategies of pest management (2)

A pest management philosophy that utilizes all suitable pest management techniques and methods to keep pest populations below economically injurious levels

Each pest management technique must be environmentally sound and compatible with producer objectives

Utilizes all suitable pest management Ways

1) Pesticide Control
2) Cultural Control
3) Mechanical Control
4) Sanitary Control
5) Natural Control
6) Biological Control
7) Host Plant Resistance

17. Physical methods names (3)

Physical Factor Controlling Insects are:

1- Heat

2-Cold

3-Moisture

4-Light

5- Electricity

18. Enlist ways of integrated pest management.(5)

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- Natural Control

- Biological Control
- Host Plant Resistance

19. Damage of rice-ear (5)

Symptoms of Damage

- The pest is essentially diurnal with highest activities in the morning and in the evening, seeking shelter during hotter parts of day
- Many generations are completed in a year
- Sucking of the grain sap causes empty or partially filled and chaffy grains and enhances subsequent fungal and bacterial infection.
- Adult and nymph both suck the sap of developing rice grains at the milking stage and cause considerable yield loss

20. Enlist plants required pollination (3)

Some plants, as the Smyrna fig, clovers, melons, tomatoes, beans, peas and most of the common fruits, require the visits of insects before the seeds or fruits can form.

21. What is scientific classification.

Scientific Classification is a system used to classify all living things through a breakdown starting with the largest grouping called a Kingdom And continuing down to the smallest grouping called species

22. Write family and order of armyworm.

Armyworm /Tobacco Cutworm:
Spodoptera litura
Family: Noctuidae
Order: Lepidoptera.
It is a major and important chewing insect pest of cotton crop.

23. What are tracheal gills.

Insects which in their early stage of live-in water cannot, of course, breathe air into their bodies through spiracles during that period of their live There are enclosed in such cases and the animal obtains and usually through special structures called tracheal gills.

24. Define vector.

Vector

An agent that transports a microorganism from one host to another. Insects that transmit plant disease are called vectors.

25. Intelligent cultivation.

Intelligent cultivation and use of fertilizers are well as pruning to remove injured or diseased parts will aid in insect control

26. Write the name of thorax segments.

The thorax has its three segments

1. Prothorax
2. Mesothorax
3. metathorax

27. Write the damage symptoms of fruit borer.

Symptoms of damage

- Bolls showing regular, circular bore holes
- Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside
- Presence of granular faecal pellets outside the bore hole.
- A single larva can damage 30-40 bolls.

28. Write the uses of pesticides.

Important aspect to use pesticides:

Must monitor pest populations in the field.

Identify the pest

Compare pest population and the economic threshold

Life stage susceptible to pesticide

Crop stage and preventable loss

29. what are repellents? Explain some repellents.

Repellents

Repellents are those materials used to keep insects away from crops, animals and man

- Various coal-tar and pine products are rather general insect repellents.
- Naphthalene is a clothes moth repellent and oil of citronella is a common ingredient of preparations used to keep mosquitoes away from people

30. What is legislative control? (2 marks)

Various laws and regulations both Federal and state have been enacted to prevent the introduction of foreign pests to prevent their spread within the country

To enforce control and extermination

To ensure that chemicals used for controlling insects are neither adulterated nor misbranded

These laws and regulations take the form of absolute quarantines, the inspection of plants to be shipped

31. Two examples of Arachnida. 2 marks

The Arachnida, including the scorpions, pseudo-scorpions, spiders, mites, ticks, etc.

32. Host plant resistance. 3 marks

Manipulating the crop to withstand or tolerate pests by

1- Natural breeding method
2- Genetically modified plants

Examples: Glandular-haired Alfalfa, Bt Corn,

33. Biological control of white grub.

Biological Methods of white grub

Bacillus papillae (Bacteria)

Beauveria bassiana (Fungus)